Supplementary material for Observe, Imagine, Abstain: Detecting Hallucinated Geometry in Feed-Forward 3D Reconstruction

Anonymous CVPR submission

Figure and table numbers below are local to this document; the main paper is self-contained and refers to this material by section title.

Reproducibility. Every number in the paper maps to a CSV and a configuration file in the released log; the benchmark is 700 lines of Python with no training, and all its model runs fit on one 16 GB GPU in about eight hours. The support head trains in about 25 minutes on one GPU from the released configuration; its runs are logged under the P3 run ids.

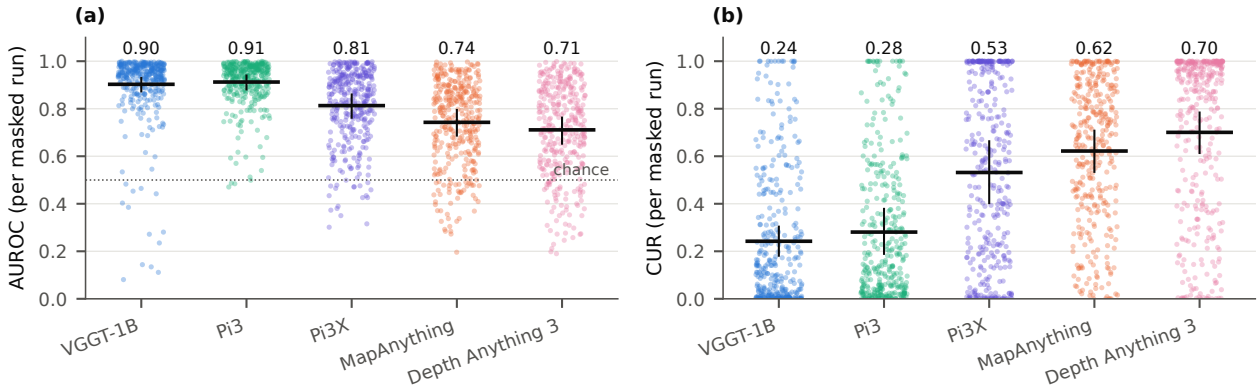


Figure S1. **Per-run spread on DTU.** AUROC and CUR for every (scan, seed, radius) run under blur and noise fills, with pooled means and intervals. The spread across scans is wide for every model; the intervals in the main tables come from resampling scans.

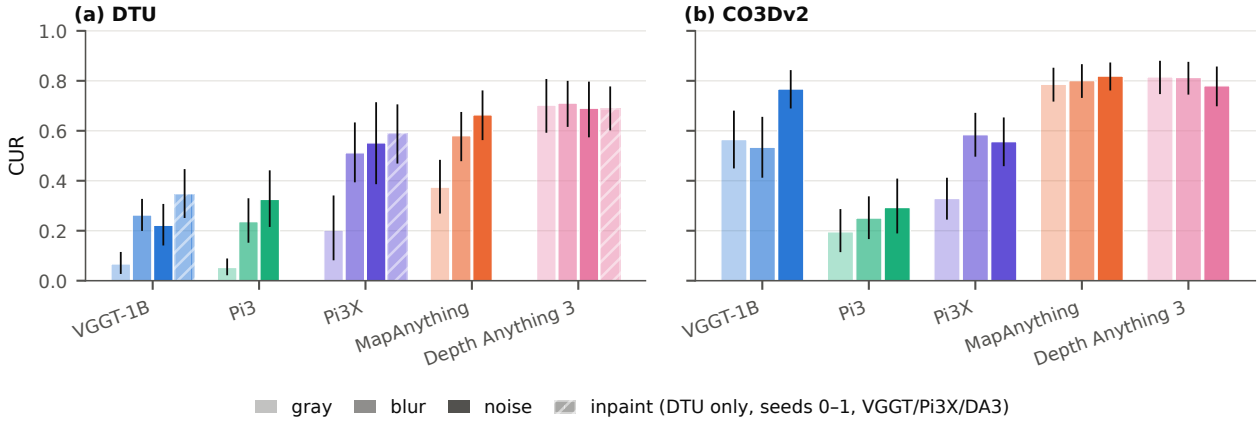


Figure S2. **Fill ablation.** CUR per model and fill on DTU and CO3Dv2 (five seeds) and for the inpainted fill on DTU (two seeds, three models). A visible gray patch gives the lowest leakage; natural-statistics and inpainted fills give more.

Correctness guards. Before any model is scored we check that ground-truth depth reprojects consistently between view pairs and that unsupported points of view i, where visible in view j, land inside view j's hidden mask. DTU: reprojection agreement 0.80 to 0.82 at a 1% depth tolerance, U-transfer 0.96 to 0.99. CO3Dv2 (cameras converted from PyTorch3D and checked against the official point cloud): with an occlusion-aware 3% tolerance, agreement 0.92 to 1.00 and U-transfer 0.90 to 0.98 on 17 of 18 scenes; the eighteenth scores 0.80, is kept, and removing it changes nothing (Section **??**). ETH3D: agreement 0.996 to 1.000 on all scenes; nine of twelve pass the U-transfer gate and three are excluded before any model is run.

Pipeline validation. On a synthetic DTU-format scene, an oracle that knows what is hidden scores AUROC 1.000 and CUR 0.000, and a blind guesser with flat confidence scores AUROC 0.500 and CUR 0.90 (the 10% of supported pixels dropped by τ is the expected floor). These numbers validate the code, not any model.

DTU and CO3Dv2, full configuration per fill.

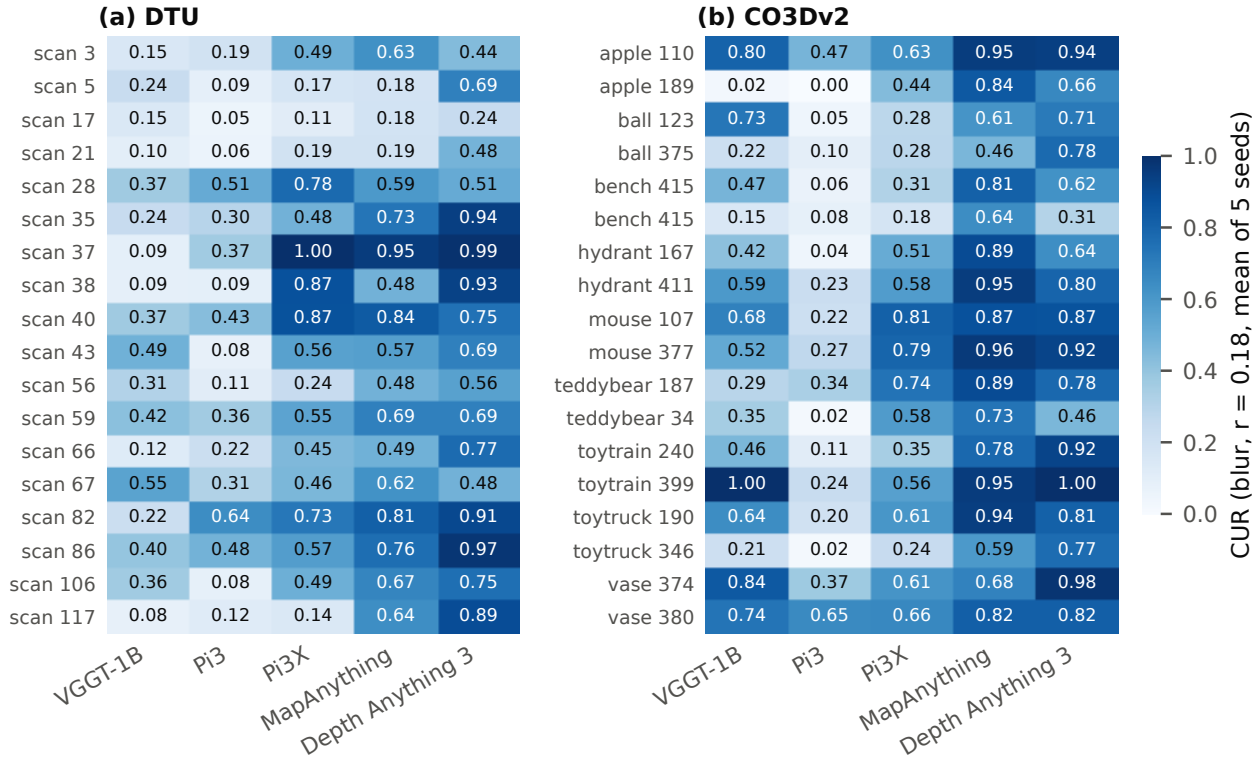


Figure S3. **Per-scene CUR** (blur fill, radius 0.18, mean over seeds) for every model on DTU scans and CO3Dv2 scenes. Rows are scenes, columns are models.

Table S1. **DTU, five models, full configuration.** 18 scans × 5 mask seeds × 2 radii × 3 fills, pooled over radii; mean [95% cluster-bootstrap CI over scans]. *conf ratio* is the median confidence inside the hidden region divided by its value under clean input. Runs P1-full and P2-full, T4, fp16. The pooled numbers are in Table 1 of the main paper.

model	fill	AUROC	CUR	CUR-ERR	AbsRel$_U$	AbsRel$_U$ ctrl	conf ratio
VGGT-1B	gray	0.973 [0.955, 0.987]	0.067 [0.031, 0.111]	0.000	0.018	0.006	0.051
VGGT-1B	blur	0.886 [0.843, 0.922]	0.263 [0.203, 0.324]	0.000	0.011	0.006	0.241
VGGT-1B	noise	0.919 [0.892, 0.945]	0.222 [0.145, 0.304]	0.000	0.014	0.006	0.192
Pi3	gray	0.981 [0.972, 0.988]	0.053 [0.025, 0.085]	0.001	0.020	0.014	0.057
Pi3	blur	0.923 [0.895, 0.947]	0.236 [0.156, 0.326]	0.001	0.018	0.014	0.261
Pi3	noise	0.902 [0.865, 0.934]	0.325 [0.219, 0.438]	0.002	0.020	0.014	0.411
Pi3X	gray	0.940 [0.902, 0.971]	0.202 [0.085, 0.337]	0.002	0.022	0.017	0.335
Pi3X	blur	0.826 [0.782, 0.865]	0.512 [0.397, 0.630]	0.002	0.020	0.017	0.739
Pi3X	noise	0.801 [0.734, 0.864]	0.551 [0.390, 0.711]	0.004	0.022	0.017	1.091
MapAnything	gray	0.861 [0.812, 0.905]	0.374 [0.272, 0.480]	0.195 [0.125, 0.269]	0.222	0.111	0.650
MapAnything	blur	0.773 [0.720, 0.825]	0.580 [0.482, 0.672]	0.277 [0.197, 0.361]	0.135	0.111	0.677
MapAnything	noise	0.712 [0.645, 0.773]	0.664 [0.566, 0.758]	0.295 [0.211, 0.390]	0.129	0.111	0.709
Depth Anything 3	gray	0.723 [0.664, 0.780]	0.703 [0.596, 0.804]	0.003	0.017	0.013	0.804
Depth Anything 3	blur	0.682 [0.611, 0.744]	0.711 [0.619, 0.796]	0.003	0.016	0.013	0.800
Depth Anything 3	noise	0.740 [0.686, 0.793]	0.690 [0.578, 0.793]	0.001	0.017	0.013	0.855

Lab rig versus phone captures, plotted.

The benchmark protocol, in a figure.

The filled patch, in pictures.

Fused point clouds at matched coverage. Each backbone's per-view depths are back-projected with its own predicted cameras into one cloud, aligned to the ground-truth frame by a similarity transform fitted on the camera centres, and compared with the ground-truth cloud of the same views at a tolerance of 1% of the scene's median depth: accuracy is the fraction of

Table S2. **CO3Dv2, five models, full configuration.** 18 scenes × 5 seeds × 2 radii × 3 fills, pooled over radii; mean [95% CI over scenes]. Runs P2-co3d and P2-co3d-b. The pooled numbers are quoted in Section 5.3 of the main paper.

model	fill	AUROC	CUR	CUR-ERR	AbsRel$_S$	AbsRel$_U$	AbsRel$_U$ ctrl	conf ratio
VGGT-1B	gray	0.747 [0.692, 0.802]	0.565 [0.453, 0.677]	0.029	0.029	0.031	0.027	0.634
VGGT-1B	blur	0.772 [0.717, 0.825]	0.534 [0.416, 0.652]	0.034	0.029	0.031	0.027	0.603
VGGT-1B	noise	0.590 [0.517, 0.656]	0.767 [0.693, 0.839]	0.029	0.029	0.029	0.027	0.805
Pi3	gray	0.930 [0.894, 0.958]	0.196 [0.118, 0.283]	0.002	0.015	0.016	0.013	0.351
Pi3	blur	0.903 [0.860, 0.939]	0.251 [0.170, 0.334]	0.002	0.015	0.016	0.013	0.389
Pi3	noise	0.880 [0.813, 0.929]	0.293 [0.193, 0.405]	0.002	0.015	0.015	0.013	0.472
Pi3X	gray	0.863 [0.825, 0.899]	0.329 [0.248, 0.408]	0.001	0.017	0.022	0.014	0.749
Pi3X	blur	0.720 [0.661, 0.777]	0.584 [0.500, 0.668]	0.002	0.017	0.018	0.014	0.877
Pi3X	noise	0.743 [0.680, 0.798]	0.556 [0.462, 0.650]	0.001	0.017	0.019	0.014	0.867
MapAnything	gray	0.609 [0.554, 0.665]	0.786 [0.721, 0.849]	0.004	0.041	0.029	0.034	0.632
MapAnything	blur	0.575 [0.514, 0.637]	0.801 [0.735, 0.863]	0.008	0.042	0.028	0.034	0.674
MapAnything	noise	0.578 [0.524, 0.631]	0.819 [0.765, 0.870]	0.007	0.041	0.033	0.034	0.690
Depth Anything 3	gray	0.619 [0.560, 0.674]	0.816 [0.751, 0.877]	0.001	0.019	0.015	0.014	0.874
Depth Anything 3	blur	0.622 [0.574, 0.668]	0.813 [0.748, 0.872]	0.001	0.018	0.015	0.014	0.853
Depth Anything 3	noise	0.618 [0.553, 0.679]	0.780 [0.702, 0.854]	0.000	0.019	0.016	0.014	0.901

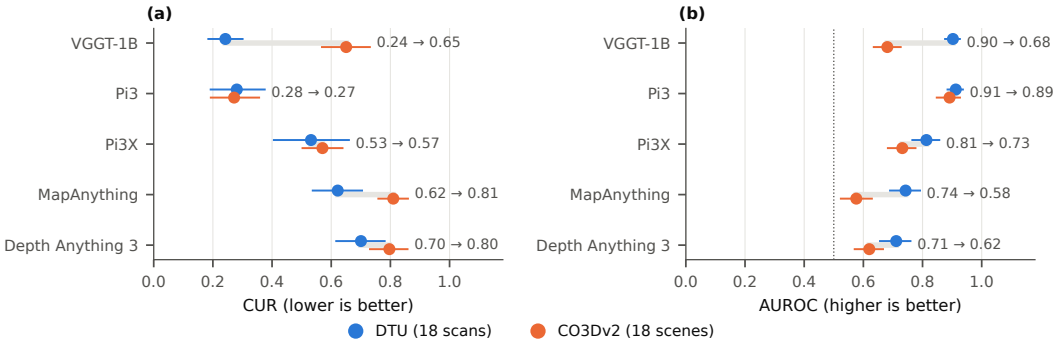


Figure S4. **Lab rig versus phone captures.** Pooled blur-and-noise CUR and AUROC per model on DTU and CO3Dv2 with 95% intervals. Pi3 is the only model whose numbers transfer; VGGT, MapAnything and DA3 leak markedly more on CO3Dv2.

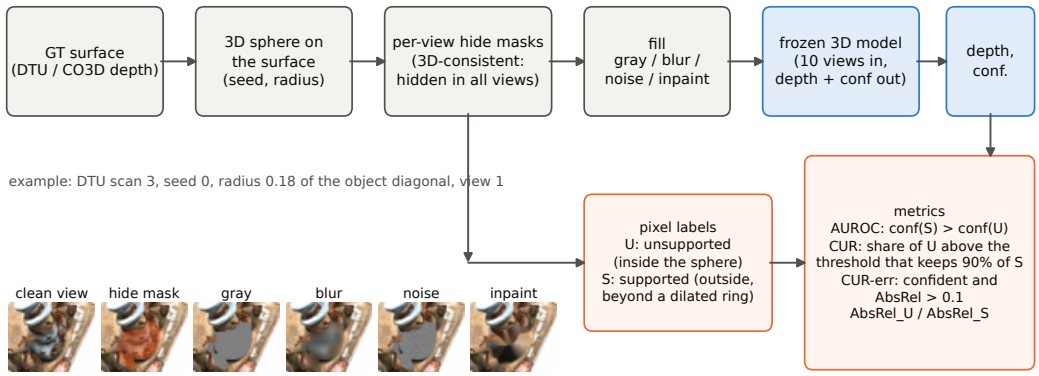


Figure S5. **The benchmark protocol.** A sphere is placed on the ground-truth surface; every pixel in every view whose ground-truth point falls inside it is hidden, so the hidden region is 3D-consistent. Hidden pixels are filled (gray, blur, noise or inpaint), the model runs on the filled views, and its depth and confidence are scored against the unsupported (U), supported (S) and ring labels. The same pixels are also scored under clean input as a control.

predicted points within tolerance of a ground-truth point, completeness the converse. A detector keeps the given fraction of points with the highest score; random dropping and an oracle that drops single-view pixels first are the references. Table S3:



Figure S6. **What a model does with a surface it cannot see.** Top: a DTU scan with one 3D region of the ground-truth surface hidden in all ten input views and inpainted so no seam remains (the inpainted fill of Section **??**). Bottom: a CO3Dv2 phone capture treated the same way. The top row shows the protocol: the same 3D sphere hides the region in every view, including a view from the other side of the rig. The bottom row shows what the backbone (VGGT) does with it: plausible geometry inside the hidden region, with confidence close to its value outside. The cyan contour marks the unsupported pixels. Model panels are the recorded run outputs.

18 DTU scans and 9 ETH3D scenes, 5 and 10 views, two view sets, run P8-fuse; the rows marked $\times$ are the label-free head multiplied by native confidence. The absolute accuracies on ETH3D are low because the predicted cameras of a building scan align poorly to the laser frame; the comparison between detectors at matched coverage is what the table is for.

Table S3. **Fused point cloud at matched coverage**, accuracy / completeness against the laser or MVS cloud.

detector	DTU, accuracy / completeness			ETH3D, accuracy / completeness		
coverage	0.6	0.8	1.0	0.6	0.8	1.0
native confidence	0.842 / 0.804	0.783 / 0.885	0.653 / 0.902	0.337 / 0.396	0.306 / 0.443	0.273 / 0.470
reprojection consistency	0.805 / 0.837	0.763 / 0.868	0.654 / 0.902	0.321 / 0.413	0.302 / 0.438	0.273 / 0.470
per-image control	0.766 / 0.735	0.698 / 0.835	0.653 / 0.902	0.293 / 0.369	0.281 / 0.428	0.273 / 0.470
label-free head	0.834 / 0.770	0.771 / 0.874	0.653 / 0.902	0.331 / 0.383	0.300 / 0.428	0.273 / 0.470
random dropping	0.653 / 0.896	0.653 / 0.900	0.653 / 0.902	0.273 / 0.462	0.273 / 0.467	0.273 / 0.470
oracle: single-view first	0.699 / 0.833	0.663 / 0.863	0.653 / 0.902	0.273 / 0.417	0.275 / 0.442	0.273 / 0.470
label-free head $\times$ native confidence	0.844 / 0.786	0.785 / 0.883	0.653 / 0.902	0.333 / 0.384	0.304 / 0.437	0.273 / 0.470
native confidence (Pi3X)	0.381 / 0.498	0.349 / 0.543	0.297 / 0.562	0.346 / 0.426	0.314 / 0.467	0.272 / 0.482
reprojection (Pi3X)	0.353 / 0.485	0.350 / 0.535	0.297 / 0.559	0.301 / 0.412	0.297 / 0.439	0.273 / 0.481
label-free head (Pi3X)	0.404 / 0.494	0.354 / 0.546	0.297 / 0.562	0.332 / 0.404	0.299 / 0.446	0.272 / 0.482
label-free head $\times$ native (Pi3X)	0.403 / 0.492	0.356 / 0.547	0.297 / 0.562	0.338 / 0.408	0.306 / 0.454	0.272 / 0.481
oracle (Pi3X)	0.310 / 0.500	0.300 / 0.531	0.297 / 0.562	0.260 / 0.418	0.268 / 0.449	0.272 / 0.482

Detectors on the filled patches. The cheap alternatives and the fill-trained heads of Section 5.4 and Section 7 of the main paper, on the Table 1 rows with blur, radius 0.18 and seed 0. A fuller perturbation baseline (photometric, blur, flip and view-permutation perturbations, $K = 8$, least-squares scale alignment, median-absolute-deviation spread; run P3-pv-full) scores AUROC 0.52 to 0.60 and CUR 0.74 to 0.86 on all five backbones over blur and noise, worse than native confidence everywhere. On 22 further DTU scans (the MVSNet test split, run P3-dtuext, seed 0, radius 0.18) the ranking of Table 1 holds: VGGT AUROC 0.908 / CUR 0.233, Pi3 0.919 / 0.272, Pi3X 0.850 / 0.539, DA3 0.755 / 0.650, MapAnything 0.697 / 0.693 with CUR-ERR 0.287. On the nine ETH3D scenes under the full fill protocol (run P3-eth3d) native confidence leaks more than on DTU, CUR 0.61 (Pi3) to 0.79 (VGGT), and the fill-trained head scores 0.981 against its per-image control's 0.987.

Stress tests. Section 5.2 of the main paper summarises this table.

Stress tests, plotted.

Table S4. **Detectors on the filled patches, DTU** (18 scans, seed 0, blur, radius 0.18; native and perturbation rows re-run on the detectors' hardware, runs P3-native and P3-pv, so every pair is scored on identical rows). The three learned rows per backbone are the *fill-trained* heads: they find the fill, not the missing support (Section **??**), which is why the per-image control matches them. The visibility-trained heads are in the main paper.

backbone	detector	fwd	AUROC	CUR
VGGT-1B	native confidence	1	0.915 [0.883, 0.943]	0.255 [0.157, 0.364]
VGGT-1B	perturbation var., K=4	4	0.666 [0.602, 0.729]	0.627 [0.521, 0.732]
VGGT-1B	reprojection consistency	1	0.488 [0.473, 0.506]	0.931 [0.903, 0.954]
VGGT-1B	fill-trained linear probe	1	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]
VGGT-1B	fill-trained per-image control	1	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]
VGGT-1B	fill-trained support head	1	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]
Pi3X	native confidence	1	0.832 [0.782, 0.879]	0.502 [0.362, 0.642]
Pi3X	perturbation var., K=4	4	0.550 [0.491, 0.608]	0.824 [0.754, 0.886]
Pi3X	reprojection consistency	1	0.478 [0.435, 0.518]	0.929 [0.899, 0.955]
Pi3X	fill-trained per-image control	1	1.000 [0.999, 1.000]	0.000 [0.000, 0.001]
Pi3X	fill-trained support head	1	0.999 [0.999, 1.000]	0.001 [0.000, 0.003]

Table S5. **Stress tests on DTU.** 18 scans × 2 mask seeds, radius 0.18, models VGGT, Pi3X, DA3. Reference rows are the matching cells of Table **??** (blur, 10 spread views, seeds 0 and 1). Last column: paired CUR difference against the reference with a sign-flip p over scans. Run P2-stress.

model	test	AUROC	CUR	CUR-ERR	AbsRel_U	CUR vs ref
VGGT-1B	reference	0.902 [0.868, 0.930]	0.259 [0.175, 0.349]	0.000	0.010	
VGGT-1B	inpaint fill	0.867 [0.821, 0.907]	0.347 [0.255, 0.443]	0.001	0.018	+0.088 ($p = 0.031$)
VGGT-1B	5 views	0.902	0.269 [0.173, 0.377]	0.000	0.011	+0.010 ($p = 0.63$)
VGGT-1B	3 views	0.899	0.259 [0.167, 0.361]	0.000	0.013	+0.001 ($p = 0.98$)
VGGT-1B	low parallax	0.911	0.214 [0.110, 0.334]	0.000	0.021	−0.045 ($p = 0.35$)
VGGT-1B	+ distractor	0.905	0.257 [0.179, 0.340]	0.000	0.011	−0.002 ($p = 0.90$)
Pi3X	reference	0.828 [0.781, 0.870]	0.521 [0.390, 0.650]	0.002	0.024	
Pi3X	inpaint fill	0.786 [0.731, 0.836]	0.589 [0.473, 0.702]	0.004	0.022	+0.068 ($p = 0.005$)
Pi3X	5 views	0.822	0.539 [0.411, 0.661]	0.006	0.024	+0.018 ($p = 0.44$)
Pi3X	3 views	0.840	0.497 [0.373, 0.618]	0.032	0.051	−0.024 ($p = 0.64$)
Pi3X	low parallax	0.812	0.523 [0.376, 0.671]	0.040	0.050	+0.002 ($p = 0.98$)
Pi3X	+ distractor	0.829	0.510 [0.368, 0.650]	0.002	0.024	−0.011 ($p = 0.66$)
DA3	reference	0.712 [0.632, 0.783]	0.679 [0.569, 0.791]	0.006	0.018	
DA3	inpaint fill	0.726	0.688 [0.605, 0.774]	0.009	0.018	+0.008 ($p = 0.87$)
DA3	5 views	0.697	0.785 [0.703, 0.865]	0.005	0.019	+0.106 ($p = 0.008$)
DA3	3 views	0.729	0.667 [0.560, 0.770]	0.018	0.031	−0.012 ($p = 0.81$)
DA3	low parallax	0.863 [0.785, 0.925]	0.372 [0.230, 0.520]	0.008	0.029	−0.307 ($p = 0.0003$)
DA3	+ distractor	0.660 [0.580, 0.734]	0.775 [0.680, 0.868]	0.003	0.019	+0.096 ($p = 0.0004$)

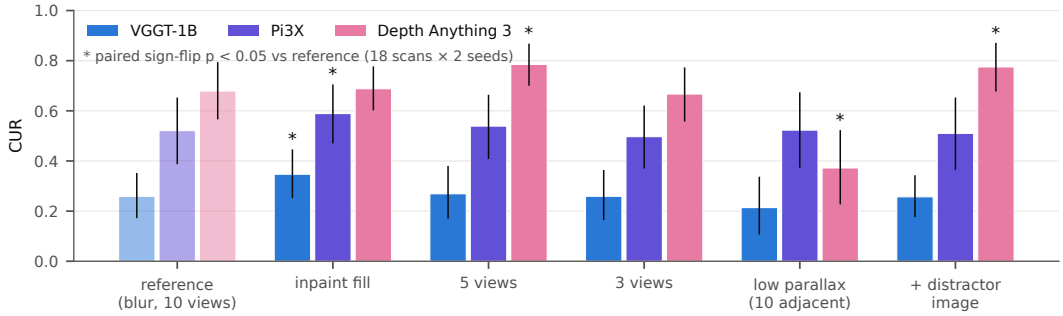


Figure S7. **Stress tests.** CUR under each stress condition against the reference, with intervals. View count does nothing for VGGT and Pi3X; the inpainted fill raises leakage; DA3 alone reacts to parallax and to a distractor.

Every detector on the true-visibility protocol. Table S6 lists every detector of the main paper's Table 3 with 95% cluster-bootstrap intervals over scenes, including the fill-trained heads and the linear probe trained on self-support labels. Per-view-count tables, the per-layer probes and every paired test are in the released statistics file (`results/P3-stats.md`).

Table S6. **True-visibility protocol, every detector**, pooled over scenes and view counts as in the main paper, with 95% cluster-bootstrap intervals over scenes.

detector	DTU		CO3Dv2		ETH3D		Photot.	
	AUROC	CUR	AUROC	CUR	AUROC	CUR	AUROC	CUR
VGGT-1B								
native confidence	0.772 [0.722, 0.823]	0.530 [0.443, 0.616]	0.754 [0.712, 0.793]	0.723 [0.650, 0.796]	0.715 [0.621, 0.802]	0.566 [0.430, 0.704]	0.802 [0.748, 0.850]	0.451 [0.343, 0.569]
reprojection consistency	0.954 [0.944, 0.963]	0.082 [0.067, 0.098]	0.793 [0.746, 0.839]	0.693 [0.517, 0.860]	0.975 [0.968, 0.982]	0.069 [0.034, 0.106]	0.824 [0.786, 0.869]	0.340 [0.249, 0.423]
distance to image border	0.701 [0.671, 0.731]	0.727 [0.700, 0.755]	0.454 [0.388, 0.520]	0.897 [0.834, 0.946]	0.725 [0.669, 0.780]	0.632 [0.534, 0.728]	0.730 [0.695, 0.765]	0.649 [0.606, 0.689]
depth edges	0.549 [0.519, 0.576]	0.831 [0.806, 0.855]	0.422 [0.373, 0.473]	0.943 [0.923, 0.962]	0.598 [0.503, 0.699]	0.762 [0.638, 0.857]	0.674 [0.594, 0.748]	0.619 [0.527, 0.714]
fill-trained head	0.609 [0.575, 0.646]	0.765 [0.728, 0.798]	0.451 [0.379, 0.522]	0.916 [0.880, 0.948]	0.686 [0.615, 0.760]	0.657 [0.551, 0.758]	0.704 [0.647, 0.759]	0.673 [0.594, 0.749]
fill-trained per-image control	0.744 [0.705, 0.779]	0.640 [0.586, 0.697]	0.356 [0.306, 0.410]	0.957 [0.943, 0.969]	0.619 [0.542, 0.698]	0.736 [0.641, 0.820]	0.622 [0.562, 0.686]	0.760 [0.706, 0.805]
per-image control (GT labels)	0.878 [0.845, 0.907]	0.340 [0.289, 0.394]	0.462 [0.392, 0.540]	0.878 [0.814, 0.933]	0.712 [0.649, 0.767]	0.666 [0.608, 0.731]	0.787 [0.732, 0.837]	0.530 [0.448, 0.603]
linear probe (GT labels)	0.989 [0.985, 0.992]	0.024 [0.015, 0.035]	0.913 [0.883, 0.942]	0.250 [0.165, 0.337]	0.966 [0.959, 0.975]	0.095 [0.068, 0.122]	0.894 [0.843, 0.936]	0.265 [0.158, 0.392]
support head, GT visibility labels	0.993 [0.990, 0.995]	0.015 [0.008, 0.024]	0.927 [0.903, 0.951]	0.211 [0.138, 0.287]	0.980 [0.972, 0.986]	0.055 [0.035, 0.076]	0.908 [0.863, 0.947]	0.258 [0.146, 0.391]
per-image control (self labels)	0.885 [0.851, 0.912]	0.325 [0.272, 0.383]	0.460 [0.394, 0.532]	0.892 [0.833, 0.937]	0.626 [0.538, 0.711]	0.750 [0.665, 0.830]	0.729 [0.656, 0.805]	0.681 [0.535, 0.810]
linear probe (self labels)	0.988 [0.984, 0.992]	0.022 [0.014, 0.032]	0.898 [0.861, 0.934]	0.296 [0.191, 0.402]	0.967 [0.958, 0.976]	0.084 [0.058, 0.107]	0.924 [0.903, 0.948]	0.187 [0.127, 0.243]
support head, label-free	0.992 [0.989, 0.995]	0.017 [0.010, 0.025]	0.931 [0.907, 0.953]	0.199 [0.134, 0.266]	0.980 [0.972, 0.986]	0.051 [0.033, 0.069]	0.930 [0.908, 0.954]	0.191 [0.125, 0.253]
label-free head, adapted on the target domain	–	–	0.841 [0.785, 0.891]	0.474 [0.338, 0.610]	–	–	0.926 [0.899, 0.952]	0.203 [0.132, 0.275]
support head (GT labels) × native confidence	0.986 [0.980, 0.990]	0.031 [0.018, 0.047]	0.920 [0.891, 0.947]	0.236 [0.149, 0.338]	0.954 [0.935, 0.972]	0.131 [0.071, 0.199]	0.931 [0.912, 0.952]	0.177 [0.115, 0.237]
label-free head × native confidence	0.980 [0.973, 0.986]	0.048 [0.030, 0.069]	0.920 [0.890, 0.948]	0.225 [0.144, 0.314]	0.943 [0.920, 0.965]	0.150 [0.086, 0.224]	0.930 [0.912, 0.951]	0.179 [0.118, 0.238]
Pi3X								
native confidence	0.820 [0.793, 0.845]	0.493 [0.435, 0.553]	0.697 [0.649, 0.740]	0.846 [0.803, 0.888]	0.887 [0.856, 0.918]	0.304 [0.213, 0.389]	0.839 [0.808, 0.867]	0.450 [0.365, 0.549]
reprojection consistency	0.914 [0.900, 0.929]	0.164 [0.120, 0.213]	0.885 [0.859, 0.909]	0.224 [0.164, 0.293]	0.977 [0.965, 0.986]	0.054 [0.024, 0.092]	0.814 [0.775, 0.857]	0.361 [0.287, 0.431]
distance to image border	0.701 [0.671, 0.731]	0.727 [0.700, 0.755]	0.454 [0.388, 0.520]	0.897 [0.834, 0.946]	0.725 [0.669, 0.780]	0.632 [0.534, 0.728]	0.730 [0.695, 0.765]	0.649 [0.606, 0.689]
depth edges	0.545 [0.516, 0.573]	0.835 [0.812, 0.858]	0.424 [0.377, 0.474]	0.951 [0.933, 0.967]	0.596 [0.498, 0.703]	0.761 [0.640, 0.854]	0.638 [0.570, 0.700]	0.699 [0.640, 0.762]
fill-trained head	0.713 [0.679, 0.745]	0.650 [0.596, 0.706]	0.433 [0.378, 0.491]	0.909 [0.864, 0.942]	0.647 [0.573, 0.703]	0.695 [0.643, 0.751]	0.823 [0.784, 0.863]	0.454 [0.370, 0.536]
fill-trained per-image control	0.718 [0.681, 0.751]	0.667 [0.622, 0.715]	0.391 [0.338, 0.445]	0.959 [0.948, 0.969]	0.644 [0.595, 0.695]	0.719 [0.625, 0.800]	0.681 [0.631, 0.730]	0.720 [0.657, 0.781]
per-image control (GT labels)	0.863 [0.831, 0.891]	0.375 [0.326, 0.425]	0.428 [0.359, 0.507]	0.889 [0.823, 0.941]	0.694 [0.640, 0.749]	0.685 [0.614, 0.745]	0.775 [0.722, 0.825]	0.577 [0.499, 0.651]
linear probe (GT labels)	0.985 [0.979, 0.989]	0.034 [0.021, 0.049]	0.931 [0.907, 0.954]	0.203 [0.133, 0.277]	0.956 [0.939, 0.970]	0.123 [0.074, 0.181]	0.919 [0.894, 0.944]	0.204 [0.136, 0.271]
support head, GT visibility labels	0.990 [0.986, 0.994]	0.021 [0.010, 0.035]	0.930 [0.907, 0.953]	0.221 [0.138, 0.312]	0.975 [0.964, 0.984]	0.072 [0.041, 0.107]	0.938 [0.919, 0.959]	0.176 [0.111, 0.240]
per-image control (self labels)	0.871 [0.837, 0.899]	0.363 [0.309, 0.418]	0.469 [0.400, 0.546]	0.877 [0.809, 0.928]	0.610 [0.531, 0.688]	0.776 [0.709, 0.831]	0.719 [0.643, 0.797]	0.697 [0.566, 0.821]
linear probe (self labels)	0.983 [0.977, 0.988]	0.036 [0.023, 0.051]	0.931 [0.903, 0.957]	0.192 [0.121, 0.269]	0.939 [0.918, 0.956]	0.171 [0.114, 0.241]	0.919 [0.894, 0.945]	0.200 [0.133, 0.264]
support head, label-free	0.988 [0.983, 0.993]	0.025 [0.013, 0.040]	0.939 [0.916, 0.960]	0.190 [0.118, 0.265]	0.963 [0.946, 0.976]	0.103 [0.061, 0.158]	0.940 [0.920, 0.960]	0.166 [0.105, 0.226]
label-free head, adapted on the target domain	–	–	0.964 [0.947, 0.978]	0.094 [0.054, 0.143]	–	–	0.929 [0.898, 0.958]	0.198 [0.118, 0.282]
support head (GT labels) × native confidence	0.988 [0.983, 0.993]	0.027 [0.014, 0.042]	0.928 [0.903, 0.952]	0.230 [0.144, 0.323]	0.978 [0.967, 0.987]	0.064 [0.033, 0.099]	0.935 [0.911, 0.957]	0.181 [0.109, 0.258]
label-free head × native confidence	0.986 [0.980, 0.991]	0.030 [0.017, 0.046]	0.935 [0.912, 0.958]	0.200 [0.124, 0.280]	0.967 [0.949, 0.980]	0.092 [0.051, 0.146]	0.934 [0.910, 0.957]	0.185 [0.113, 0.258]
Pi3								
native confidence	0.841 [0.815, 0.864]	0.489 [0.426, 0.553]	0.551 [0.503, 0.600]	0.882 [0.845, 0.917]	0.814 [0.756, 0.871]	0.498 [0.371, 0.618]	0.847 [0.795, 0.892]	0.447 [0.345, 0.555]
reprojection consistency	0.952 [0.943, 0.960]	0.089 [0.069, 0.112]	0.894 [0.871, 0.915]	0.205 [0.153, 0.261]	0.976 [0.965, 0.985]	0.039 [0.023, 0.058]	0.817 [0.775, 0.864]	0.345 [0.250, 0.432]
distance to image border	0.701 [0.671, 0.731]	0.727 [0.700, 0.755]	0.454 [0.388, 0.520]	0.897 [0.834, 0.946]	0.725 [0.669, 0.780]	0.632 [0.534, 0.728]	0.730 [0.695, 0.765]	0.649 [0.606, 0.689]
depth edges	0.546 [0.518, 0.574]	0.839 [0.816, 0.862]	0.424 [0.377, 0.472]	0.950 [0.932, 0.967]	0.597 [0.499, 0.704]	0.764 [0.637, 0.858]	0.652 [0.583, 0.718]	0.670 [0.598, 0.745]
per-image control (GT labels)	0.878 [0.846, 0.906]	0.345 [0.291, 0.402]	0.464 [0.393, 0.544]	0.880 [0.815, 0.935]	0.717 [0.651, 0.774]	0.661 [0.601, 0.729]	0.796 [0.744, 0.842]	0.510 [0.431, 0.583]
linear probe (GT labels)	0.985 [0.980, 0.990]	0.032 [0.019, 0.046]	0.943 [0.922, 0.962]	0.175 [0.109, 0.245]	0.938 [0.905, 0.965]	0.165 [0.091, 0.249]	0.914 [0.882, 0.944]	0.218 [0.143, 0.290]
support head, GT visibility labels	0.992 [0.988, 0.995]	0.016 [0.008, 0.026]	0.934 [0.914, 0.954]	0.203 [0.135, 0.273]	0.953 [0.926, 0.974]	0.135 [0.068, 0.214]	0.932 [0.911, 0.954]	0.201 [0.131, 0.272]
per-image control (self labels)	0.888 [0.856, 0.914]	0.320 [0.268, 0.381]	0.471 [0.402, 0.548]	0.895 [0.834, 0.942]	0.654 [0.579, 0.725]	0.750 [0.668, 0.822]	0.724 [0.645, 0.804]	0.683 [0.533, 0.820]
linear probe (self labels)	0.983 [0.977, 0.989]	0.031 [0.018, 0.046]	0.926 [0.897, 0.953]	0.235 [0.139, 0.335]	0.917 [0.885, 0.944]	0.225 [0.151, 0.308]	0.915 [0.882, 0.945]	0.209 [0.140, 0.275]
support head, label-free	0.989 [0.984, 0.994]	0.021 [0.011, 0.034]	0.930 [0.907, 0.952]	0.225 [0.145, 0.309]	0.951 [0.924, 0.971]	0.139 [0.075, 0.215]	0.931 [0.908, 0.955]	0.181 [0.121, 0.238]
label-free head × native confidence	0.985 [0.979, 0.990]	0.034 [0.021, 0.048]	0.915 [0.888, 0.942]	0.266 [0.172, 0.364]	0.943 [0.911, 0.966]	0.151 [0.088, 0.228]	0.929 [0.907, 0.953]	0.197 [0.129, 0.257]
Depth Anything 3								
native confidence	0.836 [0.815, 0.855]	0.430 [0.383, 0.481]	0.683 [0.615, 0.742]	0.743 [0.655, 0.828]	0.643 [0.557, 0.744]	0.635 [0.494, 0.757]	0.782 [0.707, 0.851]	0.512 [0.389, 0.636]
reprojection consistency	0.926 [0.912, 0.939]	0.175 [0.118, 0.241]	0.836 [0.799, 0.872]	0.463 [0.335, 0.585]	0.962 [0.942, 0.977]	0.109 [0.044, 0.176]	0.844 [0.804, 0.882]	0.332 [0.247, 0.419]
distance to image border	0.701 [0.671, 0.731]	0.727 [0.700, 0.755]	0.454 [0.388, 0.520]	0.897 [0.834, 0.946]	0.725 [0.669, 0.780]	0.632 [0.534, 0.728]	0.730 [0.695, 0.765]	0.649 [0.606, 0.689]
depth edges	0.551 [0.525, 0.576]	0.844 [0.821, 0.868]	0.408 [0.356, 0.463]	0.968 [0.953, 0.981]	0.592 [0.498, 0.695]	0.767 [0.643, 0.859]	0.645 [0.582, 0.707]	0.717 [0.651, 0.782]
per-image control (GT labels)	0.761 [0.728, 0.794]	0.586 [0.537, 0.632]	0.440 [0.373, 0.506]	0.903 [0.842, 0.947]	0.746 [0.701, 0.794]	0.620 [0.535, 0.700]	0.732 [0.688, 0.767]	0.646 [0.606, 0.684]
linear probe (GT labels)	0.973 [0.964, 0.980]	0.069 [0.048, 0.093]	0.875 [0.846, 0.902]	0.354 [0.283, 0.427]	0.969 [0.956, 0.980]	0.086 [0.050, 0.125]	0.909 [0.884, 0.937]	0.216 [0.153, 0.270]
support head, GT visibility labels	0.985 [0.978, 0.990]	0.036 [0.022, 0.053]	0.883 [0.859, 0.908]	0.329 [0.265, 0.392]	0.985 [0.981, 0.989]	0.040 [0.026, 0.054]	0.927 [0.900, 0.953]	0.188 [0.127, 0.244]
per-image control (self labels)	0.760 [0.727, 0.792]	0.598 [0.548, 0.647]	0.416 [0.349, 0.484]	0.911 [0.847, 0.956]	0.749 [0.700, 0.801]	0.610 [0.514, 0.700]	0.727 [0.683, 0.764]	0.650 [0.608, 0.691]
linear probe (self labels)	0.971 [0.961, 0.978]	0.074 [0.050, 0.102]	0.869 [0.836, 0.900]	0.371 [0.292, 0.451]	0.952 [0.933, 0.967]	0.134 [0.089, 0.189]	0.909 [0.886, 0.937]	0.230 [0.158, 0.284]
support head, label-free	0.982 [0.975, 0.988]	0.040 [0.026, 0.058]	0.879 [0.844, 0.911]	0.323 [0.249, 0.398]	0.974 [0.968, 0.980]	0.070 [0.049, 0.092]	0.925 [0.905, 0.948]	0.202 [0.141, 0.250]
label-free head × native confidence	0.977 [0.967, 0.984]	0.056 [0.033, 0.083]	0.833 [0.784, 0.875]	0.394 [0.302, 0.490]	0.919 [0.892, 0.947]	0.167 [0.115, 0.220]	0.921 [0.898, 0.948]	0.206 [0.139, 0.264]

Operating points from the per-pixel scores. Table S7 is computed from the per-pixel scores of run P6-dump on the labelled pixels of every cell (scene, seed, view count), each cell weighing the same. AURC: area under the single-view rate of the kept pixels against coverage (oracle: supported pixels first). AbsRel of all labelled pixels and of the half with the highest score. CUR / keep: fraction of single-view pixels passing and fraction of supported pixels kept at a threshold fixed (a) on the other scenes of the same dataset so that 90% of their supported pixels pass (leave-one-scene-out), (b) on DTU and applied unchanged (frozen; the DTU row has none). Bands: fraction of pixels marked observed / abstain / imagined at the paper's fixed thresholds; the single-view rate and AbsRel inside the observed band.

Cross-model disagreement. Two backbones run on the same views; view by view the second depth is aligned to the first by the median ratio over all valid pixels, and minus the relative disagreement is the confidence. Table S8: on the benchmark's filled patches every pair is at chance, and on real single-view geometry (DTU, clean views) disagreement scores 0.50 to 0.63, so models also agree with one another where only one view supports the surface. Agreement between models, unlike agreement between views, carries no signal in either case. Runs P8-xmodel, P8-xmodel-co3d, P8-xmodel-vis.

Fill sensitivity of the invented surface. If the surface a backbone returns inside the hidden region were fixed by the prior, changing the pixels the region is filled with would not move it. For every benchmark cell (seed 0, radius 0.18) the same hidden sphere is filled three ways (gray, blur, noise), the backbone runs on each, the three depths are aligned to the blur run by a per-view median ratio over the pixels outside the hidden region (the input mask, not a label), and every pixel's score is minus the relative spread of its depth over the three fills, so a pixel whose depth depends on the fill scores low. Table S9: AUROC and CUR of that score on the cell's U / S labels, the same for the blur-noise pair alone, the median relative spread inside and outside the hidden region, native confidence on the blur run of the same cells, and the single-view variant (the region refilled in

Table S7. **Operating points.** See the paragraph above for the columns.

dataset	backbone	detector	AURC	AbsRel all	AbsRel top half	CUR / keep (a)	CUR / keep (b)	obs. / abst. / imag.	single-view in obs.	AbsRel in obs.
DTU	VGGT-1B	native confidence	0.077	0.007	0.004	0.63 / 0.89	–	–	–	–
DTU	VGGT-1B	reprojection	0.035	0.007	0.005	0.08 / 0.94	–	–	–	–
DTU	VGGT-1B	head, GT labels	0.025	0.007	0.004	0.01 / 0.90	–	0.82 / 0.04 / 0.15	0.007	0.005
DTU	VGGT-1B	head, label-free	0.026	0.007	0.004	0.01 / 0.90	–	0.82 / 0.03 / 0.14	0.008	0.005
DTU	Pi3X	native confidence	0.057	0.021	0.015	0.59 / 0.89	–	–	–	–
DTU	Pi3X	reprojection	0.041	0.021	0.017	0.12 / 0.91	–	–	–	–
DTU	Pi3X	head, GT labels	0.026	0.021	0.014	0.01 / 0.90	–	0.79 / 0.05 / 0.16	0.006	0.016
DTU	Pi3X	head, label-free	0.026	0.021	0.014	0.02 / 0.90	–	0.77 / 0.06 / 0.17	0.005	0.016
CO3Dv2	VGGT-1B	native confidence	0.105	0.026	0.019	0.83 / 0.90	0.00 / 0.00	–	–	–
CO3Dv2	VGGT-1B	reprojection	0.088	0.026	0.021	1.00 / 1.00	0.14 / 0.53	–	–	–
CO3Dv2	VGGT-1B	head, GT labels	0.068	0.026	0.022	0.17 / 0.82	0.03 / 0.51	0.56 / 0.13 / 0.31	0.014	0.022
CO3Dv2	VGGT-1B	head, label-free	0.070	0.026	0.022	0.16 / 0.82	0.06 / 0.57	0.64 / 0.10 / 0.26	0.034	0.022
CO3Dv2	Pi3X	native confidence	0.122	0.017	0.011	0.80 / 0.84	0.83 / 0.86	–	–	–
CO3Dv2	Pi3X	reprojection	0.065	0.017	0.013	0.18 / 0.89	0.18 / 0.89	–	–	–
CO3Dv2	Pi3X	head, GT labels	0.068	0.017	0.013	0.13 / 0.83	0.06 / 0.71	0.67 / 0.13 / 0.20	0.031	0.014
CO3Dv2	Pi3X	head, label-free	0.067	0.017	0.012	0.12 / 0.84	0.06 / 0.76	0.64 / 0.16 / 0.20	0.019	0.013
ETH3D	VGGT-1B	native confidence	0.196	0.022	0.016	0.68 / 0.89	0.00 / 0.00	–	–	–
ETH3D	VGGT-1B	reprojection	0.125	0.022	0.017	0.03 / 0.93	0.03 / 0.93	–	–	–
ETH3D	VGGT-1B	head, GT labels	0.123	0.022	0.017	0.08 / 0.91	0.00 / 0.54	0.48 / 0.09 / 0.43	0.007	0.013
ETH3D	VGGT-1B	head, label-free	0.123	0.022	0.016	0.07 / 0.90	0.01 / 0.62	0.57 / 0.07 / 0.36	0.008	0.012
ETH3D	Pi3X	native confidence	0.145	0.015	0.009	0.61 / 0.90	0.62 / 0.92	–	–	–
ETH3D	Pi3X	reprojection	0.124	0.015	0.012	0.04 / 0.96	0.04 / 0.98	–	–	–
ETH3D	Pi3X	head, GT labels	0.127	0.015	0.010	0.08 / 0.90	0.01 / 0.66	0.55 / 0.13 / 0.32	0.037	0.009
ETH3D	Pi3X	head, label-free	0.129	0.015	0.010	0.13 / 0.90	0.05 / 0.79	0.60 / 0.11 / 0.29	0.052	0.009
Photot.	VGGT-1B	native confidence	0.030	0.039	0.020	0.57 / 0.89	0.00 / 0.02	–	–	–
Photot.	VGGT-1B	reprojection	0.024	0.039	0.025	0.30 / 0.91	0.30 / 0.90	–	–	–
Photot.	VGGT-1B	head, GT labels	0.013	0.039	0.026	0.27 / 0.90	0.06 / 0.59	0.67 / 0.11 / 0.22	0.007	0.026
Photot.	VGGT-1B	head, label-free	0.012	0.039	0.023	0.20 / 0.90	0.06 / 0.61	0.72 / 0.10 / 0.18	0.007	0.024
Photot.	Pi3X	native confidence	0.020	0.041	0.021	0.63 / 0.88	0.40 / 0.71	–	–	–
Photot.	Pi3X	reprojection	0.025	0.041	0.031	0.31 / 0.90	0.33 / 0.94	–	–	–
Photot.	Pi3X	head, GT labels	0.011	0.041	0.023	0.17 / 0.90	0.13 / 0.83	0.83 / 0.09 / 0.09	0.012	0.028
Photot.	Pi3X	head, label-free	0.011	0.041	0.023	0.17 / 0.90	0.19 / 0.92	0.87 / 0.06 / 0.07	0.015	0.028

Table S8. **Cross-model disagreement as a detector.** Filled patches: DTU seed 0, blur, radius 0.18 (18 scans); CO3Dv2 blur and noise, seed 0, radius 0.18 (18 scenes). Single-view: the DTU vis protocol pooled over 3, 5 and 10 views. 95% cluster-bootstrap intervals over scenes.

pair	filled patch, DTU		filled patch, CO3Dv2		single-view, DTU	
	AUROC	CUR	AUROC	CUR	AUROC	CUR
Pi3X against DA3	0.560 [0.497, 0.624]	0.831 [0.761, 0.892]	0.563 [0.513, 0.614]	0.807 [0.740, 0.869]	–	–
Pi3X against Pi3	0.574 [0.493, 0.651]	0.794 [0.710, 0.873]	0.610 [0.557, 0.660]	0.759 [0.690, 0.821]	0.592 [0.545, 0.636]	0.733 [0.687, 0.776]
VGGT-1B against DA3	0.548 [0.461, 0.631]	0.822 [0.738, 0.893]	0.545 [0.478, 0.607]	0.831 [0.769, 0.888]	–	–
VGGT-1B against MapAnything	0.613 [0.526, 0.698]	0.779 [0.651, 0.892]	0.533 [0.459, 0.606]	0.856 [0.789, 0.910]	–	–
VGGT-1B against Pi3	0.571 [0.496, 0.641]	0.781 [0.703, 0.853]	0.564 [0.494, 0.629]	0.810 [0.745, 0.871]	0.571 [0.510, 0.628]	0.746 [0.684, 0.804]
VGGT-1B against Pi3X	0.523 [0.440, 0.602]	0.865 [0.809, 0.916]	0.527 [0.452, 0.595]	0.842 [0.779, 0.900]	0.543 [0.476, 0.606]	0.765 [0.708, 0.821]

view 0 only, gray and noise against blur, scored on the pixels of that view and on those of the nine untouched views). Refilling the region in a single view moves the surface in that view (AUROC 0.93 to 0.97 on its pixels, median spread 0.7 to 5.4%) and, an order of magnitude less, in the nine untouched views (0.73 to 0.88 on theirs, spread 0.1 to 0.7%): the invented surface follows the pixels of the view it is drawn in, with a weaker pull from the other views. Run P9-refill, 18 DTU scans and 18 CO3Dv2 scenes, four backbones.

Table S9. **Fill sensitivity of the invented surface.** The depth spread over three fills of the same hidden sphere as a detector, against native confidence on the same cells; the last two columns of each block refill one view only. 95% cluster-bootstrap intervals over scenes.

backbone	DTU, 18 scans								CO3Dv2, 18 scenes							
	AUROC	CUR	AUROC (blur, noise)	spread U (%)	spread S (%)	native AUROC	one view, own	one view, others	AUROC	CUR	AUROC (blur, noise)	spread U (%)	spread S (%)	native AUROC	one view, own	one view, others
VGGT-1B	0.978 [0.965, 0.987]	0.053 [0.028, 0.083]	0.934	2.0	0.1	0.915	0.969	0.882	0.910 [0.871, 0.943]	0.209 [0.135, 0.292]	0.844	1.1	0.2	0.839	0.934	0.843
Pi3X	0.918 [0.889, 0.943]	0.225 [0.152, 0.312]	0.854	1.6	0.3	0.831	0.930	0.842	0.965 [0.952, 0.976]	0.079 [0.049, 0.117]	0.903	1.1	0.1	0.778	0.935	0.823
Pi3	0.927 [0.873, 0.962]	0.174 [0.099, 0.274]	0.876	2.1	0.3	0.928	0.938	0.754	0.947 [0.930, 0.963]	0.125 [0.080, 0.177]	0.887	0.8	0.1	0.934	0.944	0.843
Depth Anything 3	0.824 [0.767, 0.875]	0.437 [0.318, 0.563]	0.806	2.0	0.6	0.714	0.960	0.729	0.620 [0.543, 0.699]	0.723 [0.604, 0.823]	0.612	1.3	0.9	0.684	0.775	0.630

Twenty views. With 20 clean DTU views, twice the most the heads saw in training (run P4-vis-ext), the visibility heads score AUROC 0.990 to 0.993 and the label-free heads 0.990 to 0.992, native confidence falls to 0.61 (VGGT) and 0.83 (Pi3X), reprojection consistency scores 0.93 and 0.89, and the per-image control rises to 0.94: with twenty views the few remaining single-view pixels sit at the frame edge, and the control reads the edge.

Pre-registered gates. The pass conditions were written into the run registry before the corresponding heads were trained, and the registry is versioned with the code. Gate 3 (true-visibility head; written 2026-11-06, before any visibility head was trained; the training kernels ran from commit `4132d11`): the head counts as detecting missing support when, pooled over 3, 5 and 10 views on DTU, (a) it beats native confidence on AUROC and CUR (paired sign-flip $p < 0.05$), (b) it beats its per-image control on AUROC by at least 0.02 at $p < 0.05$, and (c) both conditions hold on at least one dataset the head never saw (CO3Dv2 or ETH3D); if (b) fails, the single-view-support signal is reported as recoverable from per-image features and the method claim is dropped. Gate 4 (label-free head; written 2026-11-06 16:51 UTC, before any label-free head was trained, commit `c7e6919`): on DTU and on CO3Dv2, (a) and (b) as above against its own per-image control, and (c) within 0.03 AUROC of the ground-truth-trained head on DTU and ahead of reprojection consistency on CO3Dv2 at $p < 0.05$ (its training signal is the reprojection test, so it has to generalise better than its teacher); if (a) or (b) fails the head is dropped, if only (c) fails it is reported as an ablation. The earlier Gate 2 (fill-trained head) had the clause "head ahead of the control at $p < 0.05$" with no margin; it passed by the letter on differences of 0.0004 and 0.003 and the paper does not count it, which is why Gate 3 carries a margin.

Gaussian-splatting backbone (partial run). AnySplat [**?**], a feed-forward 3D Gaussian-splatting model, was scored on the DTU configuration with its per-pixel Gaussian opacity as confidence (120 of 540 rows) and with the rendered accumulated alpha (450 of 540 rows), bf16 on one L4, run P3-gs. Pooled over blur and noise: opacity AUROC 0.583 / CUR 0.718, rendered alpha 0.632 / 0.745, CUR-ERR 0.02 and 0.06; the noise fill drives the opacity below chance (0.36). The representation does not change the answer: a Gaussian the model places on an unseen surface is as opaque as one on an observed surface. The run stopped behind other jobs on the shared card and is reported as partial.

Run registry. Every table cell maps to a run id (P1-full, P2-full, P2-stress, P2-co3d, P2-co3d-b, P3-pv-smoke, P2-smoke-b, P3-native, P3-reproj, P3-pv, P3-oia-dtu, P3-oia-abl, P3-oia-stress-*, P3-kaggle-vggt) whose CSV, configuration JSON and kernel log are released with the code, together with the dated log of every failed run and every correction.